ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2021-2022**

DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4305: Computer Organization and Architecture

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all **6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

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1. a) It is a common practice to use a symbolic representation of machine instructions to write a machine-language program. Consider the following high-level language instruction: 2+2+3+8 (CO2) (PO1)

$$X = \frac{(A + B \times C)}{D - E \times F} \times (G + \frac{H}{I})$$

i. What are the common sources of these operands?
ii. What are the basic elements of a machine instruction?
iii. Draw an instruction cycle state diagram showing all the steps involved in instruction execution
iv. Write two separate programs using two- and one-address machine instruction formats respectively to compute the above expression. Values of given operands should not be changed during execution.

b) A *procedure* is the most significant invention in the evolution of programming languages. It is an independent computer program that is embedded within a bigger program. The procedure may be invoked or called at any time within the program. An example program where procedures are called multiple times during its execution is illustrated in Figure 1. 5+5 (CO3) (PO2)

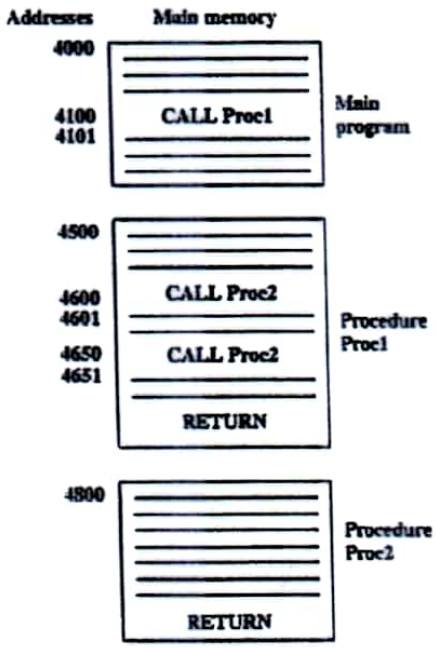


Figure 1: A program with *procedure calls*.

i. Draw an execution sequence diagram illustrating how the procedures are called at different addresses during program execution.
ii. Based on the given scenario, demonstrate how a processor employs a stack data structure to store return addresses and implement nested procedure calls.

2. In a uniprogramming system, the main memory is divided into two sections: one for the operating system and one for the currently executing program. The later section of this multiprogramming system's memory is partitioned to accommodate multiple processes. Suppose a process references five pages, A, B, C, D, and E, and the following sequence of virtual page numbers is observed on a computer with virtual memory:

A; B; C; D; A; B; E; A; B; C; D; E

a) What is process? Why is its state considered important in multiprogramming? Draw a figure illustrating the life cycle of a process. — 5 (CO3) (PO2)

b) Why would the main memory of a multiprogramming system such as the preceding example be expanded hypothetically? Discuss briefly the common methods utilized by operating systems to increase main memory with required figures. — 10 (CO3) (PO2)

c) Employ each replacement algorithm to compute the total number of page transfers during this sequence of references, beginning with an empty main memory that contains four-page frames. Demonstrate the intermediate phases of the calculation, taking into account any necessary assumptions. — 10 (CO3) (PO2)

3. a) In a typical instruction style, the address field(s) are quite small. However, we prefer to access a large number of locations in main memory or virtual memory. Various addressing strategies have been utilized to achieve this objective. — 2+5+5+8 (CO2) (PO1)

   i. How does the processor determine the addressing mode of an instruction's operand(s)?
   ii. In displacement addressing mode, how does the CPU calculate the effective address (EA)? Which variations does this mode entail? Introduce them briefly.
   iii. Compare post- and pre-indexing addressing mode with the relevant figures.
   iv. If an instruction's address field contains the decimal value 14, determine the EA of the corresponding operand for all conceivable basic addressing modes. Assume a mandatory variable in any field that requires one and also discuss any anomalies, if any.

   b) Various techniques can be used to predict whether a branch will be taken or not. Among them dynamic branch prediction approaches depend on the execution history. Briefly explain those approaches with appropriate figures. — 5 (CO5) (PO1)

4. a) The ALU of a computer system can add two input registers and logically complement the bits of either register, but it doesn't have any built-in logic to perform subtraction. The integers must be stored in twos complement format in those registers. Assume R1 and R2 are the addresses of those registers. — 4+5+5+6 (CO5) (PO1)

   i. Write a program of machine instructions to perform subtraction operation between these two registers.
   ii. Draw the data flow diagram (*i.e.*, a sequence of events) to perform the same subtraction operation.
   iii. Make a list the micro-operations that the associated control unit must do in order to reformulate the event sequence you have previously generated.
   iv. If the registers contain 11110000 and 0010100 respectively what would be the value of the following flags: Carry, Zero, Overflow, Sign, Even Parity, Half-Carry

   b) Explain *"cycle stealing"* with necessary figure. — 5 (CO6) (PO1)

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5. a) To the concept of instruction execution, we must recognize that, in fact, an instruction has a number of stages. Such a pipelined processor has a clock rate of 2.5 GHz and executes a program with 1.5 million instructions. The pipeline has six stages, and instructions are issued at a rate of one per clock cycle. 4+3+3+5 (CO5) (PO1)

   i. Draw a flow chart of six stage "*CPU instruction pipeline*" depicting the logics necessary for branches and interrupts
   ii. Ignoring penalties due to branch instructions and out-of-sequence executions what is the speedup of this processor for this program compared to a nonpipelined processor? Show the intermediate calculating steps.
   iii. What is the pipelined processor's throughput (in MIPS) with the same assumptions?
   iv. What is pipeline bubble? Provide some instances demonstrating how this event can occur in a variety of ways under the aforementioned circumstances, along with the required figures.

   b) Assume a disk drive comprises of four drives with a 200 GB capacity each. 4+6 (CO4) (PO1)

   i. Draw a diagram illustrating the mapping between logical and physical disk space for such disk drive.
   ii. What is the available data storage capacity for each of the RAID levels 0, 1, 3, 4, 5, and 6?

6. a) The issue with programmed I/O is that the CPU must wait a considerable amount of time for the I/O module in question to be prepared for data receiving or transmission. A different method for the processor to accept interrupts from I/O modules in order to complete such data communications. 2+6+6+6 (CO6) (PO1)

   i. What role does the I/O module play in this strategy?
   ii. The occurrence of an interrupt initiates a series of hardware and software processor activities. With an appropriate flow diagram explain how such interrupt is processed to complete an I/O operation in this way.
   iii. Implementing Interrupt-driven I/O entailed a number of design issues. Discuss these concerns and their variants briefly.
   iv. What is *interrupt controller*? Draw a diagram of a computer system illustrating how I/O devices can be connected to a processor using interrupt controller(s).

   b) The DMA mechanism can be configured in numerous ways to regulate bus access. Design some potential configurations for this aspect. 5 (CO6) (PO1)